Chapter 8. Main memory. Exercises

Original problems. State assumptions, units, and intermediate reasoning. Answers are in the separate solutions PDF.

OS08-E01. Page translation

Page size is 4 KiB. Virtual address 0x2A7C has page number 2. The entry maps page 2 to resident frame 9 with read-only permission.

1. Compute offset and physical address.
2. What happens on a write?
3. If the entry is not currently usable, must that imply a design error?

Check your approach. Retain offset width and permission checks throughout the arithmetic.

OS08-E02. Effective access time

A serialized TLB lookup costs 2 ns and each memory access 100 ns. A TLB miss requires one additional page-table memory access. Assume equal memory costs, no other cache or page-walk cache, and no page faults in the first two parts. Let h=0.95.

1. Derive EMAT and evaluate it.
2. Recompute at h=0.80.
3. How should page-fault cost enter the model?

Check your approach. Enumerate the actual hit and miss operations before weighting them.

OS08-E03. Page-size tradeoffs

There are 100 objects of 33 KiB each. Each object has its own page-aligned allocation and cannot share its tail page with another object. Compare 4 KiB pages with a 64 KiB alternative.

1. How much tail space does one 4 KiB-page allocation waste?
2. Why can 64 KiB pages reduce entry count?
3. Calculate 64 KiB-page tail waste per object and for all 100. Are smaller pages always more memory-efficient overall? Distinguish space from translation and fault-service time.

Check your approach. Include both space and translation costs, with their allocation assumptions.

OS08-E04 An unfamiliar x86-64 walk

Use x86-64, four levels, 4 KiB pages and 48-bit canonical VA; all accesses are at CPL=3 (user mode). The CPU supports NX and IA32_EFER.NXE=1; SMEP, SMAP, PKE and other extra mechanisms do not intervene. CR3=0x7000; PDPT/PD/PT bases are 0x8000/0x9000/0xA000. VA 0x00001234_5678_9ABC maps frame 0x12AB. Every level is present, user-accessible and RW, with NX=0. After changing a PTE, complete the necessary TLB invalidation and synchronization on every CPU that might cache the mapping before testing it.

1. Check canonical and give all four indices and offset.
2. Use 8-byte entries to find entry addresses and plausible values.
3. Derive PA.
4. Keep the same mapping but set PTE 0x12AB005. Give read/write results.

OS08-E05 Canonical, present, and permissions

Use E04's mode, CPL=3, NXE=1 and permission assumptions. Each case starts independently; after changing an entry, complete affected CPUs' TLB invalidation and synchronization. Test (A) VA 0x00002468_ACE0_1234; (B) E04's canonical VA with its PML4E.P=0; (C) E04's original mapping with PTE.NX=1; (D) VA 0x00008000_00000000. A's entries are not specified; do not assume it takes E04's path.

1. Give A's indices/offset and canonical result.
2. Which case rejects before walking and which faults at a table level?
3. Which data reads/writes succeed and which fetches fault?
4. Can a present-only check distinguish write protection and NX? Explain entry bit 63 when NXE=0 and give a branch table.

OS08-E06 ARM layout, domain, and AP

Use ARMv7-A AArch32 short descriptors, TTBCR.N=0 and PD0=0, SCTLR.AFE=0 and TRE=0, 4 KiB small pages, PL0 accesses and DACR.D3=Client. P1 root is 0x14000 and L2 is 0x1C000; P2 root is 0x18000 and L2 is 0x1C400. L1 raw words are 0x1C061/0x1C461 and both L2 words are 0x54862. P1 VA is 0x004A6000 and P2 VA is 0x007C8000. The processes have distinct, unreused ASIDs and exactly these two mappings to frame 0x54000, with no other mappings, pins or lifetime references. Track the mapping count, initially 2, not a general implementation's complete refcount. After unmapping, complete required TLB invalidation, barriers and synchronization on all affected cores before testing or reclaiming.

1. Prove storage sizes/alignment/nonoverlap before finding L1/L2 indices and four entry addresses.
2. Decode all raw fields, using bit 9 and bits 5:4 for small-page AP. Give both processes' read/write results.
3. Remove only P1's L2 entry and complete TLB maintenance. Give P2 read, P1 read and mapping-count results. When can the frame be reclaimed?
4. Change D3 to Manager or AFE to 1. Does the original write-fault conclusion still hold?

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz, Galvin and Gagne, Operating System Concepts, 9th ed., Chapter 8. Intel 64 and IA-32 SDM, Vol. 3A; Arm Architecture Reference Manual ARMv7-A/R.

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https://github.com/xinhi6716/computer-science-self-study